

Documentation Guideline Manual

> **Version:** 1.0.0 · **Status:** Canonical · **Owner:** Documentation Team

> A single source of truth for writing, structuring, and maintaining project documentation.

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1. Philosophy & Core Principles

Good documentation is a product, not an afterthought. It has users, versions, and a maintenance lifecycle. Apply the same rigour to docs as you do to code.

The Four Qualities

Quality	What it means
Clear	A first-time reader understands it without asking clarifying questions
Accurate	Every claim, code sample, and screenshot reflects the current state of the software
Minimal	Nothing present that doesn't serve the reader's goal
Findable	A reader who doesn't know the exact term can still locate the content

The Reader-First Rule

Before writing a single word, answer three questions:

1. **Who** is reading this? (new contributor, end user, API integrator, ops engineer)
2. **What** do they need to accomplish?

3. **What do they already know?**

Write for the reader you identified, not for yourself.

Documentation Debt

Undocumented features and outdated docs are technical debt. Treat them as bugs:

- Outdated content → file an issue with label `docs: stale`
- Missing content → file an issue with label `docs: missing`
- A PR that changes behaviour **must** update the relevant docs in the same PR

2. File & Folder Conventions

Naming Rules

...

✓	getting-started.md	kebab-case, lowercase
✓	api-reference.md	
✓	CONTRIBUTING.md	SCREAMING-SNAKE for repo root conventions
✓	README.md	
✓	CHANGELOG.md	
✗	GettingStarted.md	No PascalCase
✗	getting_started.md	No underscores
✗	doc1.md	No meaningless names
✗	new-doc-FINAL-v3.md	No version suffixes in filenames

Standard Repo Root Files

Every repository **must** contain these files at the root:

File	Purpose
`README.md`	Project overview, quickstart, and navigation hub
`CHANGELOG.md`	Versioned history of user-facing changes
`CONTRIBUTING.md`	How to contribute – code, docs, issues
`LICENSE`	Legal licence (no extension)
`.github/PULL_REQUEST_TEMPLATE.md`	PR checklist for contributors
`.github/ISSUE_TEMPLATE/`	Templates for bug reports and feature requests

Recommended Docs Folder Structure

...

docs/	
├─ index.md	# Documentation home / overview

```

├── getting-started/
│   ├── installation.md
│   ├── quickstart.md
│   └── configuration.md
├── guides/
│   ├── authentication.md
│   └── deployment.md
├── api/
│   ├── overview.md
│   ├── endpoints.md
│   └── errors.md
├── reference/
│   ├── cli.md
│   └── environment-variables.md
├── contributing/
│   ├── development-setup.md
│   ├── code-style.md
│   └── documentation.md
...

```

← this manual lives here

3. Document Structure

Heading Hierarchy

Use headings to express logical structure, not visual style.

```markdown

# H1 – Document title. One per file. Never repeated.

## H2 – Major section. Appears in table of contents.

### H3 – Subsection.

#### H4 – Use sparingly. If you need H5, consider splitting the document.

```

Rules:

- Never skip heading levels (H1 → H3 with no H2 is invalid)
- Never use bold text as a substitute for a heading
- Every document has exactly **one H1**, which matches its filename intent
- Headings are sentence case: `## Getting started` not `## Getting Started`

Required Document Sections

Every documentation page should open with:

```markdown

# Page Title

Brief one- or two-sentence description of what this page covers and who it is for.

---  
```

Long pages should include a **Table of Contents** immediately after the opening description.

Section Order Convention

For feature/guide pages:

1. **Overview** – what this feature does and why it exists
2. **Prerequisites** – what the reader needs before starting
3. **Steps / How to use** – the core content
4. **Examples** – concrete, copy-pasteable samples
5. **Configuration reference** – all available options in a table
6. **Troubleshooting** – common errors and fixes
7. **Related** – links to adjacent docs

Not every page needs every section. Include what's relevant.

4. Markdown Syntax Reference

Emphasis

````markdown``  
**Bold** – strong importance, warnings, key terms (first use)  
*Italic* – titles of works, technical terms, mild emphasis  
~~Strikethrough~~ – deprecated options, removed features  
``inline code`` – all code, commands, filenames, env vars, and UI strings  
`````

> **Rule:** Never use bold for decoration. Use it to signal that the reader **must** pay attention.

Lists

Unordered – use when order doesn't matter:

````markdown``  
- Item one  
- Item two  
  - Nested item (two spaces before dash)  
  - Another nested item  
- Item three  
`````

Ordered – use for sequential steps:

```
```markdown
1. Clone the repository.
2. Install dependencies.
3. Start the development server.
```
```

****Rules:****

- Use ``-`` for unordered list bullets (not ``\*`` or ``+``)
- Always number ordered lists ``1.``, ``2.``, ``3.`` – not ``1.``, ``1.``, ``1.``
- End each list item with a period if any item is a full sentence
- No period if items are fragments (labels, single words, short phrases)
- Nest a maximum of two levels; beyond that, restructure

Tables

```
```markdown
| Column A | Column B | Column C |
|-----|-----|-----|
| Value | Value | Value |
```
```

****Alignment:****

```
```markdown
| Left-aligned | Center-aligned | Right-aligned |
|:-----|:-----:|-----:|
| text | text | text |
```
```

****Rules:****

- Every table has a header row
- Align numeric columns right
- Keep cell content short – if a cell needs a paragraph, use a definition list instead
- Never use a table just to display key-value pairs with two columns; use a definition list or bold+colon inline

Code Blocks

Always specify the language for syntax highlighting:

```
```markdown
```bash
npm install
```
```

```
```python
```

```
def greet(name: str) -> str:
    return f"Hello, {name}"
...
```

```
```json
{
 "version": "1.0.0",
 "debug": false
}
...
....
```

**\*\*Common language identifiers:\*\***

Language	Identifier
Bash / Shell	`bash` or `sh`
JavaScript	`javascript` or `js`
TypeScript	`typescript` or `ts`
Python	`python`
JSON	`json`
YAML	`yaml`
SQL	`sql`
HTML	`html`
CSS	`css`
Dockerfile	`dockerfile`
Plain text / output	`text`
File tree / structure	`text`

**\*\*Rules:\*\***

- Inline code (`` `backticks` ``) for short fragments in prose
- Fenced blocks (`` ``` ``) for anything spanning multiple lines
- Always include a language identifier – never a bare `` ``` ``
- Include only the minimum code needed to illustrate the point
- Add comments inside the block when a line needs explanation

---

### ### Blockquotes

Use for callouts, notes, and warnings:

```
```markdown
> **Note:** This feature requires Node.js 18 or later.

> **Warning:** Running this command will permanently delete data.

> **Tip:** You can skip this step if you already have Docker installed.
...`
```

Rendered convention (use consistently throughout the project):

> **Note:** Informational. The reader should know this but can proceed without acting.

> **Warning:** The reader must act on this or risk data loss, security issues, or breakage.

> **Tip:** Optional improvement or shortcut.

> **Deprecated:** This feature will be removed. Use the linked alternative.

Horizontal Rules

Use `---` (three dashes) on its own line to divide major sections in long documents. Do not use `\*\*\*` or `\_\_\_`.

Images

```markdown

![Alt text describing the image](path/to/image.png)

<!-- With a title tooltip: -->

![Dashboard screenshot](assets/dashboard.png "Ebola Data Explorer dashboard")  
```

Rules:

- Alt text is **mandatory** – describe what the image shows, not what it represents
- Alt text for a screenshot: describe what's visible on screen
- Alt text for a diagram: summarise the information it conveys
- Store images in `docs/assets/` or `assets/` relative to the doc
- Use PNG for screenshots and diagrams; SVG for icons and logos

5. Writing Style Guide

Voice & Tone

|  Do |  Don't |

|-----|-----|

| Write in second person: "You can configure..." | First person plural: "We recommend..." |

| Active voice: "The CLI creates a config file" | Passive voice: "A config file is created" |

| Present tense: "The function returns..." | Future tense: "The function will return..." |

| Direct: "Run `npm install`" | Indirect: "You might want to consider running..." |
| Confident: "This approach is faster" | Hedged: "This approach might possibly be faster" |

Sentence & Paragraph Length

- Maximum **one idea per sentence**
- Maximum **4-5 sentences per paragraph**
- If a paragraph exceeds 6 lines, find the break
- Short sentences are not unprofessional – they are easier to translate and scan

Terminology Consistency

Pick one term and use it everywhere. Document your choices:

```
```markdown
<!-- In a project glossary or contributing guide: -->
| Preferred term | Do not use |
|-----|-----|
| user | end-user, consumer, client |
| repository | repo (in formal docs), project |
| configuration | config (acceptable in code/CLI context) |
| authenticate | log in (for human auth flows only) |
```
```

Numbers & Units

- Write out numbers one through nine: "three attempts", "nine retries"
- Use numerals for 10 and above: "10 retries", "256 MB"
- Always include the unit: "5 seconds", "2 GB", "100 ms"
- Use a space between the number and unit: `100 MB` not `100MB`

Dates & Times

- ISO 8601 for timestamps: `2024-08-08T00:00:00Z`
- Human-readable dates: `August 8, 2024` (not `8/8/24` or `08-08-2024`)
- Always specify timezone when time matters: `14:00 UTC`

Avoid These Words

| Word | Why | Alternative |
|-----------------------|---------------------------------------|---|
| simply / just / easy | Condescending if the reader struggles | Remove the word |
| obviously / clearly | Same | Remove the word |
| very / really / quite | Filler | Remove the word |
| etc. | Vague | List everything, or write "and similar" |
| Note that... | Padding | Start with the note itself |
| Please | Unnecessary in technical docs | Remove the word |
| leverage | Jargon | Use / utilise |
| seamless | Marketing | Be specific about what's easy |

6. Code Documentation

Inline Comments

Write comments that explain **why**, not **what**. The code explains what.

```
```python
```

```
 Bad - describes what the code does (obvious from the code)
x = x + 1 # increment x
```

```
 Good - explains why
retry_count += 1 # WHO API returns 429 on burst; back off before retry
```
```

Function / Method Docstrings

Python (Google style):

```
```python
```

```
def calculate_cfr(cases: int, deaths: int) -> float:
 """Calculates case fatality rate as a percentage.
```

Args:

cases: Total confirmed case count. Must be greater than zero.  
deaths: Total confirmed death count. Must be <= cases.

Returns:

CFR as a percentage (0.0-100.0).

Raises:

ValueError: If cases is zero or deaths exceeds cases.

Example:

```
>>> calculate_cfr(28616, 11310)
39.52
```

```
"""
...
```

**JavaScript / TypeScript (JSDoc):**

```
```javascript
```

```
/**
```

* Calculates the Wilson Score confidence interval for a proportion.

*

* @param {number} k - Number of positive outcomes (deaths).

* @param {number} n - Total trials (cases). Must be > 0.

* @param {number} [confidence=0.95] - Confidence level (0-1).

* @returns {{ estimate: number, lower: number, upper: number }}

* CFR estimate and 95% CI bounds, all as percentages.

* @throws {RangeError} If n is zero or k exceeds n.

*

```

* @example
* wilsonCI(11310, 28616);
* // → { estimate: 39.52, lower: 39.02, upper: 40.03 }
*/
function wilsonCI(k, n, confidence = 0.95) { ... }
```

```

### ### README for Code Files

Any source file with complex logic should open with a brief comment block:

```

```python
"""
ebola_model.py
-----
Simplified SIR (Susceptible-Infected-Recovered) model for the
2014-2016 West Africa Ebola outbreak.

```

Usage:

```
python ebola_model.py --R0 1.7 --gamma 0.067 --days 730
```

References:

```

    Camacho et al. (2014), Lancet Infectious Diseases
    Agua-Agum et al. (2015), NEJM

```

```

"""
```

```

```

```

### ## 7. API Documentation

#### ### Endpoint Documentation Template

Each endpoint entry must contain:

```

```markdown
## `GET /v1/outbreaks/{id}`

```

Returns detailed data for a single Ebola outbreak record.

Path Parameters

Parameter	Type	Required	Description
`id`	string		Outbreak identifier (e.g. `2014-west-africa`)

Query Parameters

Parameter	Type	Default	Description
`fields`	string	all	Comma-separated list of fields to return
`format`	enum	`json`	Response format: `json` or `csv`

Request Headers

Header	Required	Description
`Authorization`	✓	`Bearer <token>`
`Accept`	✗	`application/json` (default)

Example Request

```
```bash
curl -X GET "https://api.example.com/v1/outbreaks/2014-west-africa" \
 -H "Authorization: Bearer YOUR_TOKEN" \
 -H "Accept: application/json"
```
```

Example Response – 200 OK

```
```json
{
 "id": "2014-west-africa",
 "cases": 28616,
 "deaths": 11310,
 "cfr": 39.5,
 "start": "2013-12-01",
 "end": "2016-06-09",
 "countries": ["GIN", "LBR", "SLE"]
}
```
```

Error Responses

| Status | Code | Description |
|--------|----------------|---|
| `400` | `INVALID_ID` | The outbreak ID is malformed |
| `401` | `UNAUTHORIZED` | Missing or invalid token |
| `404` | `NOT_FOUND` | No outbreak found with this ID |
| `429` | `RATE_LIMITED` | Exceeded 100 req/min – retry after `Retry-After` header |

Error Code Documentation

All error codes must be documented in a single `errors.md` reference:

```
```markdown
`NOT_FOUND`

HTTP Status: 404
Meaning: The requested resource does not exist or has been deleted.
Resolution: Verify the ID is correct. Check the resource list endpoint.
```
```

8. README Standards

Every project's `README.md` must follow this structure:

```markdown

# Project Name

One sentence: what this project does and who it is for.

[![License: MIT](badge)] [![Version](badge)] [![Build](badge)]

> Optional: a one-line quote, key stat, or value proposition.

---

### ## Features

- Brief bullet list of the top 4-6 capabilities
- Use present tense: "Exports to CSV" not "Can export to CSV"

### ## Quick Start

The fastest path from zero to running. Maximum 5 steps.  
If it takes more, there's a setup problem to fix.

### ## Installation

Detailed installation for all supported environments.

### ## Usage

Most common use case, fully worked example with real output.

### ## Configuration

Table of all environment variables / config options.

### ## Project Structure

File tree with one-line explanation of each top-level entry.

### ## Contributing

Link to CONTRIBUTING.md. One-paragraph summary of the process.

### ## License

[MIT] (LICENSE) — © 2024 Your Name  
```

Badge Standards

Use shields.io for badges. Place them immediately after the title:

```
```markdown
[![License: MIT](https://img.shields.io/badge/License-MIT-yellow.svg)](https://opensource.org/licenses/MIT)
[![npm version](https://img.shields.io/npm/v/package-name)](https://www.npmjs.com/package/package-name)
[![Build Status](https://img.shields.io/github/actions/workflow/status/org/repo/ci.yml)](...)
```
```

****Rules:****

- Maximum 6 badges
- Only link to meaningful destinations
- No vanity badges (star count, visitor count)

9. Changelog Format

Follow [Keep a Changelog](https://keepachangelog.com) exactly.

```
```markdown
Changelog
```

All notable changes to this project will be documented in this file.

The format follows [Keep a Changelog](https://keepachangelog.com/en/1.1.0/) and this project adheres to [Semantic Versioning](https://semver.org/spec/v2.0.0.html).

---

### ## [Unreleased]

#### ### Added

- CFR Calculator now supports custom confidence levels (80%, 90%, 99%)

#### ### Changed

- District Profiler sort order now defaults to CFR descending

---

### ## [2.1.0] – 2024-11-07

#### ### Added

- HCW Explorer country-breakdown view with per-country CFR comparison
- Keyboard shortcut `R` resets Outbreak Replay to frame 0

#### ### Changed

- Epidemic curve now defaults to monthly-new view instead of cumulative

### ### Fixed

- Timeline scrubber no longer skips frame 0 on drag (#142)
- Monrovia node label overlaps Margibi at 1× zoom (#138)

### ### Deprecated

- `calculateCFR(n, k)` argument order – swap to `(cases, deaths)` before v3.0

---

## [2.0.0] – 2024-08-08

### ### Added

- What If? Simulator with three-parameter counterfactual model

### ### Removed

- Legacy table-only dashboard (replaced by interactive chart dashboard)

### ### Security

- Upgraded Chart.js from 4.3.0 to 4.4.0 (CVE-2024-XXXX)

---

## [1.0.0] – 2024-03-01

Initial public release.

[Unreleased]: <https://github.com/org/repo/compare/v2.1.0...HEAD>

[2.1.0]: <https://github.com/org/repo/compare/v2.0.0...v2.1.0>

[2.0.0]: <https://github.com/org/repo/compare/v1.0.0...v2.0.0>

[1.0.0]: <https://github.com/org/repo/releases/tag/v1.0.0>

``

### ### Change Categories (in order)

Category	Use for
`Added`	New features
`Changed`	Changes to existing functionality
`Deprecated`	Soon-to-be removed features
`Removed`	Now-removed features
`Fixed`	Bug fixes – include issue number
`Security`	CVE fixes, dependency upgrades for security

### \*\*Rules:\*\*

- Write from the **user's perspective** – what changed for them, not what you changed in the code
- One line per change; if it needs more, link to a GitHub issue or PR
- Keep `[Unreleased]` at the top – move it to a version heading on release
- Never delete old entries – the full history must be preserved

---

## ## 10. Diagrams & Visuals

### ### When to Use a Diagram

Use a diagram when the prose equivalent would require more than a paragraph to explain a relationship or flow.

Diagram type	Use for
Flowchart	Decision trees, process flows, conditional logic
Sequence diagram	API calls, event sequences, auth flows
Entity diagram	Data models, database schemas
Architecture diagram	System components and their relationships
State diagram	Lifecycle of an entity (order states, connection states)

### ### Mermaid Diagrams (Preferred)

Use [Mermaid](https://mermaid.js.org) for diagrams that can be version-controlled as text. GitHub renders Mermaid natively in Markdown files.

```
```markdown
```mermaid
graph TD
 A([User submits case count]) --> B{cases > 0?}
 B -- No --> C[/Show validation error/]
 B -- Yes --> D[Calculate CFR]
 D --> E[Compute Wilson CI]
 E --> F[/Render result + chart/]
...
```
```

```
```mermaid
graph TD
 A([User submits case count]) --> B{cases > 0?}
 B -- No --> C[/Show validation error/]
 B -- Yes --> D[Calculate CFR]
 D --> E[Compute Wilson CI]
 E --> F[/Render result + chart/]
...
```
```

****Common Mermaid diagram types:****

```
```markdown
<!-- Sequence diagram -->
```mermaid
sequenceDiagram
    Client->>API: GET /outbreaks/2014
    API->>Cache: lookup
    Cache-->>API: miss
    API->>DB: query
```
```

```
DB-->>-API: row
API-->>-Client: 200 OK { ... }
...
```

```
<!-- Entity relationship -->
```mermaid
erDiagram
    OUTBREAK ||--o{ COUNTRY : affects
    OUTBREAK {
        string id PK
        int cases
        int deaths
        date start
    }
    COUNTRY {
        string iso PK
        string name
    }
    ...
    ...
```

Screenshot Standards

- Capture at ****1440×900**** viewport minimum
- Use light mode unless documenting a dark-mode feature
- Crop to the relevant UI element – not the full browser window
- Annotate with arrows or boxes only when a specific element is not obvious
- File format: PNG, max 500 KB; compress with [Squoosh](https://squoosh.app) if larger
- Filename: `feature-name-context.png` (e.g. `cfr-calculator-ci-bar.png`)

11. Links & Cross-references

Link Syntax

```
```markdown
<!-- Inline link – preferred for most uses -->
Read the [installation guide](../getting-started/installation.md).

<!-- Reference link – preferred when the same URL appears 3+ times -->
See the [WHO situation reports][who-reports] for primary source data.

[who-reports]: https://www.who.int/emergencies/situations/ebola-outbreak-2014-2016-West-Africa
```
```

Link Text Rules

```
```markdown
❌ Click [here](url) for the guide.
❌ See [this page](url) for details.
```

✗ [https://example.com/docs/guide](https://example.com/docs/guide)

✓ Read the [configuration guide](url).

✓ See [Semantic Versioning](url) for the full specification.

Link text must describe the destination – never use "here", "this", "link", or a raw URL.

### ### Internal vs External

- Internal links: use **relative paths** (`../api/endpoints.md`), not absolute URLs
- External links: use absolute URLs with `https://`
- Do not link to specific GitHub branch URLs (they break) – link to the file in the default branch

### ### Anchor Links

Generate heading anchors by converting the heading to lowercase, replacing spaces with `-`, and removing punctuation:

```
```markdown
## API Reference & Errors
<!-- Anchor: #api-reference--errors -->

[Jump to errors](#api-reference--errors)
```
```

---

## ## 12. Accessibility

Documentation is read by people using screen readers, translation tools, and low-vision assistive technology. Apply these rules unconditionally.

### ### Alt Text

```
```markdown
<!-- ✗ Empty alt text on a meaningful image -->
![ ](chart.png)

<!-- ✗ Filename as alt text -->
![chart.png](chart.png)

<!-- ✗ "Image of..." – screen readers already announce it's an image -->
![Image of the CFR bar chart](chart.png)

<!-- ✓ Describes the content -->
![Bar chart comparing CFR across eight Ebola outbreaks from 1976 to 2022, with 1976 Yambuku highest at 88.1% and 2014 West Africa lowest at 39.5%](chart.png)
```
```

...

### ### Colour & Contrast

- Never convey information by colour alone – add a label, icon, or pattern
- Severity levels: do not write "the red districts are most affected" – write "districts rated EXTREME"
- Charts: use distinguishable patterns or direct labels in addition to colour

### ### Table Headers

Every table must have a header row:

```
```markdown
| Outbreak | Cases | Deaths | CFR |
|-----|-----|-----|-----|
| ...      | ...   | ...     | ...   |
```
```

Screen readers use the header row to contextualise cell content.

### ### Reading Level

- Target a reading level appropriate to your audience
- Avoid unnecessary jargon; define technical terms on first use
- Use sentence case (not title case) for headings – easier to scan and translate

---

## ## 13. Templates

### ### 13.1 Feature Guide Template

```
```markdown
# Feature Name
```

Brief description: what this feature does, who uses it, and why it matters.

Prerequisites

- Requirement one
- Requirement two

Overview

One paragraph explaining the conceptual model before the how-to steps.

Steps

1. **Step title.** Description of what to do and why.

```
```bash
command --flag value
```
```

Expected output:

```
```text
Success: thing created.
```
```

2. **Step title.** Continue...

Configuration Options

| Option | Type | Default | Description |
|---------|---------|---------|--|
| name | string | - | Required. Human-readable identifier. |
| timeout | integer | 30 | Seconds before the operation is cancelled. |

Examples

Example: Basic usage

```
```bash
tool run --name "my-run"
```
```

Example: With custom timeout

```
```bash
tool run --name "my-run" --timeout 60
```
```

Troubleshooting

Error: `CONNECTION_REFUSED`

Cause: The server is not running.

Fix: Start the server with `tool server start` before running this command.

Related

- [Configuration reference](../reference/configuration.md)
- [Authentication guide](authentication.md)

13.2 Bug Report Template

```
```markdown
```

```

name: Bug report
about: Something isn't working as documented
labels: bug

Description

A clear, one-sentence description of the bug.

Steps to Reproduce

1. Go to '...'
2. Click on '...'
3. Observe '...'

Expected Behaviour

What you expected to happen.

Actual Behaviour

What actually happened. Include error messages verbatim.

Environment

| Field | Value |
|-----|-----|
| OS | macOS 14.4 |
| Browser | Chrome 124 |
| Version | v2.1.0 |

Screenshots / Logs

<!-- Attach if relevant -->
````
```

13.3 Pull Request Template

```
``markdown
## Summary
```

One sentence describing what this PR does and why.

Closes #____

Type of Change

- ☐ Bug fix (non-breaking)
- ☐ New feature (non-breaking)
- ☐ Breaking change
- ☐ Documentation update only

- [] Refactor / performance (no behaviour change)

Documentation

- [] I updated the relevant documentation in the same PR

- [] No documentation changes needed (explain below)

Testing

- [] I tested this change manually

- [] I added or updated automated tests

Screenshots (if UI change)

<!-- Before / After -->

Checklist

- [] Code follows the project style guide

- [] Self-review completed

- [] No console.log / debug code left in

- [] CHANGELOG.md updated under `[Unreleased]`

```\n

---

## ### 13.4 Architecture Decision Record (ADR) Template

```\nmarkdown

ADR-001: Use Chart.js Instead of D3 for Charts

Date: 2024-08-01

Status: Accepted

Deciders: @alice, @bob

Context

We need interactive charts across five tools. Options considered:
D3.js, Chart.js, Recharts, Plotly.

Decision

Use Chart.js 4.x for the four standalone HTML tools. Use Recharts for the React dashboard (alignment with existing component model).

Rationale

- Chart.js is available as a single UMD CDN bundle – no build step
- Covers all chart types needed (line, area, bar, horizontal bar)
- Smaller runtime footprint than Plotly
- Recharts not suitable outside React; D3 requires significantly more code

Consequences

****Positive:**** No build pipeline required for standalone apps.
****Negative:**** Two charting libraries in the project; contributors must know both.
****Mitigation:**** `data/ebola-data.js` is shared; chart code is isolated per app.

Alternatives Considered

| Library | Rejected Because |
|----------|---|
| D3.js | Too much low-level code for standard chart types |
| Plotly | 3.4 MB bundle; overkill for bar/line charts |
| Recharts | Requires React; not suitable for vanilla HTML files |

14. Review & Maintenance

Documentation Review Checklist

Before merging any PR that touches documentation:

...

Structure

- ☐ Correct heading hierarchy (no skipped levels)
- ☐ One H1 per file, matches the page intent
- ☐ Table of contents present on pages > 300 words

Content

- ☐ Accurate – tested against current software behaviour
- ☐ No outdated screenshots (check version numbers, UI elements)
- ☐ All code samples execute without error
- ☐ Links resolve (no 404s)

Style

- ☐ Second person, active voice, present tense
- ☐ No banned words (simply, just, obviously, etc.)
- ☐ Consistent terminology (matches project glossary)

Accessibility

- ☐ All images have meaningful alt text
- ☐ No colour-only information conveyance
- ☐ All tables have header rows

Markdown

- ☐ Language specified on all code blocks
- ☐ No trailing whitespace
- ☐ File ends with a single newline

...

Staleness Policy

| Trigger | Required Action |
|--------------------------|--|
| Feature added or changed | Update docs in the same PR |
| API endpoint modified | Update `api/endpoints.md` and `CHANGELOG.md` |
| Quarterly review | Audit all docs for accuracy; file issues for stale content |
| Major version release | Full documentation audit before release tag |

Ownership

Each documentation file has an owner. Assign owners in a `CODEOWNERS` file:

...

```
# .github/CODEOWNERS
```

```
# Documentation
```

```
docs/api/           @api-team
docs/guides/        @product-team
docs/contributing/ @dx-team
README.md           @project-lead
CHANGELOG.md        @release-manager
```

...

15. Tooling

Recommended Tools

| Tool | Purpose | Install |
|-----------------------|---|--|
| [markdownlint] | (https://github.com/DavidAnson/markdownlint) | Lint Markdown for style errors `npm i -g markdownlint-cli` |
| [Vale] | (https://vale.sh) | Prose linter (style, tone, terminology) `brew install vale` |
| [markdown-link-check] | (https://github.com/tcort/markdown-link-check) | Validate all links in .md files `npm i -g markdown-link-check` |
| [Prettier] | (https://prettier.io) | Auto-format Markdown consistently `npm i -D prettier` |
| [Mermaid Live] | (https://mermaid.live) | Preview Mermaid diagrams in browser web app |

Markdownlint Configuration

Save as `.markdownlint.json` at the repo root:

```
```json
{
 "default": true,
```

```

 "MD013": { "line_length": 120 },
 "MD024": { "siblings_only": true },
 "MD033": false,
 "MD041": true
 }
}

```

### ### Vale Configuration

Save as ``.vale.ini`` at the repo root:

```

```ini
StylesPath = .vale/styles
MinAlertLevel = suggestion

[*.*md]
BasedOnStyles = Vale, write-good
```

```

### ### Pre-commit Hook

Add to ``.pre-commit-config.yaml`` to enforce standards before every commit:

```

```yaml
repos:
  - repo: https://github.com/igorshubovych/markdownlint-cli
    rev: v0.41.0
    hooks:
      - id: markdownlint
        args: ["--config", ".markdownlint.json"]

  - repo: https://github.com/tcort/markdown-link-check
    rev: v3.11.2
    hooks:
      - id: markdown-link-check
        args: ["--quiet"]
```

```

### ### CI Integration

Add to ``.github/workflows/docs.yml``:

```

```yaml
name: Docs Quality

on: [push, pull_request]

jobs:
  lint:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4

```

```

- name: Lint Markdown
  uses: DavidAnson/markdownlint-cli2-action@v16
  with:
    globs: "**/*.md"

- name: Check links
  uses: gaurav-nelson/github-action-markdown-link-check@v1
  with:
    use-quiet-mode: yes
    use-verbose-mode: yes
...

```

Appendix A – Glossary

Term	Definition
ADR	Architecture Decision Record – a short document capturing a significant technical decision, its context, and its rationale
Canonical	The single authoritative version of a document; others should link to it rather than copy it
Docstring	A documentation comment embedded in source code, adjacent to the function or class it describes
Frontmatter	YAML metadata block at the top of a Markdown file, delimited by <code>---</code> , used by static site generators
Keep a Changelog	A changelog format convention at keepachangelog.com that this project follows
Semver	Semantic Versioning – version numbers in <code>MAJOR.MINOR.PATCH</code> format with defined rules for incrementing each part
Stale	Documentation that describes a past state of the software and has not been updated to reflect current behaviour

Appendix B – Quick Reference Card

...

HEADINGS	# H1 (one per file) · ## H2 · ### H3 · #### H4 (sparingly)
BOLD	text – importance, warnings, key terms (first use only)
ITALIC	<i>text</i> – titles, technical terms, mild emphasis
CODE (inline)	<code>command</code> – all code, filenames, env vars, UI strings
CODE (block)	<pre>language ...</pre> – always specify language
LISTS	- unordered 1. ordered (never * or +)
LINKS	[text](url) – never "click here" or raw URLs
IMAGES	![alt text](path) – alt text is mandatory
TABLES	always include header row; right-align numeric columns
CALLOUTS	> Note: / > Warning: / > Tip: / > Deprecated:
RULES	--- (three dashes, own line)
VOICE	second person · active · present tense · direct

AVOID simply · just · obviously · clearly · etc. · leverage ·
seamless
STRUCTURE Overview → Prerequisites → Steps → Examples → Config →
Troubleshooting
```

---

\*This manual is a living document. File a `docs: improvement` issue or  
open a PR to propose changes. All updates follow the same review process  
described in [Section 14](#14-review--maintenance).\*

---

**\*\*Document metadata\*\***

| Field         | Value              |
|---------------|--------------------|
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| Created       | 2024-01-01         |
| Last reviewed | 2024-11-07         |
| Owner         | Documentation Team |
| License       | CC BY 4.0          |